ADARSH KUMAR

Delhi, India | +91 9315847748 | adarshkumar2024@gmail.com

Linkedin

PROFESSIONAL SUMMARY

Systems-oriented BCA student specializing in **Industrial IoT (IIoT) logic** and **Backend Data Pipelines**. Expert in architecting simulation environments for **Industry 4.0**, with a focus on **Predictive Analytics** and **Automated Quality Control**. Dedicated to supporting Lean Manufacturing through the development of intelligent monitoring systems and risk-classification algorithms. Committed to the *Kaizen* philosophy of continuous digital optimization to eliminate *Muda* (waste) in automotive production.

TECHNICAL CORE

- **Data Pipelines & Backend:** Python (Flask), SQL (Schema Design for IIoT), API Integration.
- **Predictive & Risk Logic:** Rule-based Scoring Engines, Risk Classification, Logic-based Automation.
- **Industrial Monitoring:** Production Dashboard Simulation, Real-time KPI Tracking (OEE), OCR Data Digitization.
- **Tools:** GitHub (Version Control), MySQL (Historical Data Archiving), VS Code.

INDUSTRIAL SIMULATION & PROJECTS

Smart Factory Monitoring System (Industry 4.0 Architecture)

Lead Developer | Python, SQLAlchemy, Flask

- **Predictive Analytics:** Engineered a backend pipeline that simulates real-time production data to calculate **OEE (Overall Equipment Effectiveness)**.
- **Production Monitoring:** Developed a dynamic dashboard for live tracking of machine uptime, downtime, and throughput, modeled on automotive assembly line requirements.
- **Automated Quality Control:** Integrated logic-based "Fail-Safe" triggers to simulate *Jidoka* (Autonomation), flagging defect patterns before they reach final assembly.
- **Data Archiving:** Designed a structured MySQL schema for historical performance logging, enabling root-cause analysis of simulated production bottlenecks.

Intelligent Risk Classification Engine (Security & Compliance)

Lead Developer | Python, OCR Integration

- **Risk Logic:** Built a multi-tier classification engine utilizing keyword-based risk scoring to automate the identification of anomalies.
- **Digital Transformation:** Integrated OCR to convert analog/image-based reports into structured digital datasets, supporting the digitization of manual factory logs.

Scalable Industrial Web Architecture

Systems Architect

- Designed a modular backend architecture to support high-concurrency data requests, simulating the infrastructure needed for a plant-wide **Manufacturing Execution System (MES)**.